

Table 1.

Instrument specifications including gauge type, make and model, location and field campaign, wind shield type (if applicable), deployment period, sampling interval, reported accuracy, spatial resolution (if applicable), and known measurement issues with relevant references. Asterisks (*) denote instruments requiring external power for heating or data logging.

Instrument	Make & Model	Location(s) & Campaign	Wind Shield	Deployment Period	Sampling Interval	Reported Accuracy	Spatial Resolution	Known Issues
In-Situ Precipitation Gauges								
Storage Gauge <i>[Reference]</i>	—	BB; Butte SNOTEL; Schofield Pass SNOTEL	Single-Alter	Ongoing	10 min	—	—	Snow capping over orifice; wind-driven undercatch (Serreze et al., 1999; McGurk, 1986)
Weighing Bucket*	OTT Pluvio2	GT (SAIL); KP (SPLASH)	SDFIR (GT); LPDF (KP)	Sep 2021–Jun 2023	1 min; 30 min†	1%	—	Wind-driven undercatch (Rasmussen et al., 2012); wetting losses via evaporation or sublimation (Groisman & Legates, 1994; Savina et al., 2012); snow capping; burial of LPDF wind shield in deep snowpack (Kochendorfer et al., 2023); power-dependent heating.
Tipping Bucket*	Novalynx 260-2500E-12	GT (SAIL)	SDFIR	Sep 2021–Jun 2023	1 min	3% liquid	—	Filter clogging and riming during heavy snowfall cause severe undercatch (Nitu & Wong, 2010; Kochendorfer et al., 2022); wind-driven undercatch; snowmelt timing errors during mixed-phase events.
Optical Rain Gauge*	Optical Scientific ORG-815-DS	GT (SAIL)	—	Sep 2021–Jun 2023	1 min	5% liquid; 10% solid	—	Underperforms during solid precipitation and low precipitation rates (Nystuen et al., 1996; Kyrouac & Tuftedal, 2024); falsely registers blowing snow as precipitation (Fassnacht, 2004; Cappozi et al., 2021); sensitive to particle size and shape (Kochendorfer et al., 2022; Smith et al., 2020).
Present Weather Detector*	Vaisala PWD22	GT (SAIL)	—	Sep 2021–Jun 2023	1 min	Not applicable	—	Reduced accuracy during frozen precipitation; classification errors during mixed-phase events; susceptible to blowing snow contamination (Kyrouac & Tuftedal, 2024).
Laser Disdrometer*	OTT Parsivel2	GT (SAIL); KP (SPLASH)	—	Sep 2021–Jun 2023 (GT); Sep 2023– (KP)	1 min	5% liquid; 20% solid	—	Assumes liquid water density; overestimates snowfall without ρ–D correction (Boudala et al., 2014); blowing snow particles falsely registered as precipitation; ρ–D adjustment uncertainty from snow crystal habit variability (Heymsfield et al., 2004; Brandes et al., 2007; Holroyd, 1971); wind speed reduces apparent fall velocity, affecting precipitation rate estimates.
Snow Pillow	Alpine Hydromet FSP5 Fluidless	KP (SOS)	—	Oct 2022–Jun 2023	5 min	0.3% (SWE only)	—	Measures SWE only; not designed for event-scale precipitation timing (2015); wind redistribution of snow complicates accumulation signal (Lundquist et al., 2024); valid only when fully snow-covered.
Gridded Precipitation Products								
SQUIRE	CSU X-Band Radar (SAIL)	Gridcell at GT & KP (SAIL)	—	Dec 2021–Mar 2022; Dec 2022–Mar 2023	5 min	—	250 m	Z–S relationship uncertainty varies with snow crystal habit and storm type (Matrosov et al., 2009; Jackson et al., 2026); winter months only; underperforms at high precipitation rates; ensemble spread reflects Z–S uncertainty.
PRISM	—	Gridcell at GT & KP	—	Oct 2021–Sep 2023	1 day	—	800 m	Reference gauge incorporated in PRISM algorithm, inflating apparent agreement (Daly et al., 2008, 2021); sparse station network in complex terrain limits performance for small events (Henn et al., 2018).
ERA5-Land	ECMWF	Gridcell at GT & KP	—	Oct 2021–Sep 2023	1 hr	—	~9 km	Coarse resolution cannot resolve valley-scale precipitation variability; parameterized convection struggles with localized warm-season convection in complex terrain (Beck et al., 2019).
ASO SWE	Airborne Snow Observatories	3×3 km area (BB, GT, KP)	—	1 Apr 2023	Instantaneous	—	50 m	Single flight limits temporal coverage; snow density derived from modeled values rather than direct measurement (Painter et al., 2016); used here as estimate of accumulated winter precipitation only.

* Instrument requires external power for heating elements or data logging. † At GT, the "not-real-time" 30-min variable was used for the weighing bucket per manufacturer recommendation, as it undergoes additional proprietary processing to improve accuracy (Bartholomew, 2020). BB = billy barr site; GT = Gothic Town site; KP = Kettle Ponds site; SDFIR = Small Double Fence Intercomparison Reference; LPDF = Low-Porosity Double Fence; SWE = snow water equivalent; Z–S = radar reflectivity–snowfall rate relationship; ρ–D = snow density–diameter relationship.